
Portable Predictions: Learning Housing Prices Across Diverse Markets

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Abstract

Accurate housing price forecasting cannot be achieved with structural attributes alone. Portable Predictions is a computer platform that incorporates ACS PUMS data, ZIP-level spatial joins, and California county-level crime data to improve home value estimation. Using models such as linear regression, Ridge, random forest, and XGBoost, we find that adding violent and property crime rates significantly improves R^2 and reduces bias. The architecture integrates SHAP-based explainability, automated reporting, residual diagnosis, and investment scoring. Our results highlight the importance of geographic enrichment and model explainability to ensure scalable and reliable housing prediction tools.

1 Introduction

Precise housing price forecasting continues to be a key challenge for real estate agents, city planners, policymakers, and mortgage lenders. Sound estimates assist in determining affordability, inform investment decisions, as well as manage mortgage risk on the local and regional levels. Although traditional statistical models have long supported this research, modern machine learning (ML) techniques - and ensemble techniques in particular, like Random Forest and XGBoost — offer greater flexibility in capturing the complex, nonlinear relationships that define housing markets in diverse geographies Sharma et al. [2024], Ho et al. [2020], Mora-García et al. [2022].

However, generalization is a prominent limitation of current Automated Valuation Models (AVMs). Most are trained on data from a single city and do not account for neighborhood-level variation in characteristics such as crime, school quality, and the composition of the housing stock, characteristics known to influence the value of houses. As El Jaouhari and Samadhiya [2024] point out, this limits the reliability of AVMs when used in new environments.

In this regard, our Portable Predictions platform combines structural and socioeconomic features from the ACS Public Use Microdata Sample (PUMS) with ZIP code-level crosswalks and California crime data integrations. We benchmark a set of baseline and models - linear regression, Ridge regression, Random Forest, and XGBoost — and leverage SHAP values to investigate the importance of the features and the limitations of the model.

Preliminary results show that models that use structural features only underperform in areas with limited context. Incorporating crime data has improved residual balance and model confidence, particularly within high-variance counties. We address current sources of prediction error in incomplete geographic joins between neighborhoods and neighborhood feature gaps and outline a plan for future model enhancement through improved spatial integration and feature enrichment.

Our final goal is to provide a usable and understandable tool that transmits localized housing price data to support decision-making for a range of users.

2 Related Work

Housing price modeling has long used linear regression as a benchmark due to its transparency and interpretability. However, real-world housing markets exhibit nonlinear behavior due to structural, socioeconomic, and geographic factors - requiring increasingly more accommodative models. Ridge regression adds robustness to linear models by penalizing multicollinearity, but ensemble methods such as Random Forest and XGBoost have been consistently shown to have superior predictive power in varied datasets when coupled with well-designed features Sharma et al. [2024], Ho et al. [2020].

During the COVID-19 pandemic, abrupt market shocks put the limitations of fixed models in the limelight. Mora-García et al. [2022] illustrated that the increasing algorithms outdid traditional econometric models by handling nonlinear, high-variance market patterns more successfully Mora-García et al. [2022]. However, even with advances within the model configuration, geographic generalization remains a significant constraint. El Jaouhari and Samadhiya [2024] found that most Automated Valuation Models (AVMs) are learned from expert data sets and do not generalize between zones with different zoning, demographic, or environmental conditions El Jaouhari and Samadhiya [2024].

To address this, more recent research has attempted to add spatial context to prediction models. Derdouri and Murayama [2020] proposed Regression-Kriging to represent local geographic heterogeneity, and Li [2023] created GeoShapley to integrate predictive accuracy and spatial attribution Derdouri and Murayama [2020], Li [2023]. These are especially useful for real estate applications, where location-based attributes such as crime and school quality can change dramatically within a few meters.

Interpretability is one area that has received attention in housing models. Tree-based methods work well, but their "black box" nature might discourage uptake in high-stakes settings. SHAP (SHapley Additive exPlanations) has also become a standard method for explaining model outputs. Liu [2024] followed up by combining SHAP with geospatial visualizations to highlight local variation of importance of the features. Others emphasize the role of transparency in building trust, particularly in settings such as lending or public investment Li [2025].

Our effort builds on these strands by integrating structural and socioeconomic data from ACS PUMS with geographic crosswalks and crime data at the county level for the state of California. We evaluated baseline and ensemble models (Linear, Ridge, Random Forest, XGBoost) on a shared pipeline that includes SHAP for explanation and assess measures of portability across counties. By progressively incorporating contextual data and tracking performance region by region, we aim to identify where core AVMs fall short, and how spatial characteristics can improve accuracy and credibility.

3 Methodology

Our approach builds a reproducible end-to-end pipeline for housing price prediction that integrates household-level socioeconomic features, engineered housing attributes, and neighborhood-level crime context. Following best practices in interpretable machine learning for real estate Sharma et al. [2024], Ho et al. [2020], Mora-García et al. [2022], we combine microdata from the American Community Survey (ACS PUMS) with geographic crosswalks at the ZIP code level and California crime statistics to enhance spatial grounding.

We evaluated four core models: Linear Regression, Ridge Regression, Random Forest, and XGBoost. These models are trained to predict logarithmic transformation of housing values ($\log(\text{VALP} + 1)$) to stabilize variance and improve residual behavior. For linear models, we apply feature scaling via `StandardScaler`; tree-based models use raw features due to their scale invariance.

3.1 Modeling Framework

Let $X \in \mathbb{R}^{n \times d}$ denote the matrix of housing and household characteristics and $y \in \mathbb{R}^n$ the reported property values. To address right-skewed distributions and stabilize variance, we transform prices using a log function.

$$y^{(\log)} = \log(y + 1).$$

We evaluated four predictive models drawn from both linear and ensemble-based families:

1. **Linear Regression:** A transparent baseline model fit via Ordinary Least Squares.
2. **Ridge Regression:** A regularized variant using an L_2 penalty to mitigate multicollinearity.
3. **Random Forest:** An ensemble of decision trees trained by bootstrap aggregation to capture non-linear feature interactions.
4. **XGBoost:** A gradient-boosted tree model optimized for tabular data and strong out-of-the-box performance.

Linear and Ridge models are trained on standardized features using `StandardScaler`, while tree-based models use raw, unscaled input due to their scale-invariant nature.

The performance of the model is assessed on a test set canceled using the root mean square error (RMSE) and the coefficient of determination (R^2) on the logarithmic transformed target. At this stage, hyperparameter tuning is limited to default settings or shallow grid search, providing a baseline for future optimization.

3.2 Data Sources and Feature Design

Our primary data source is the 2023 ACS Public Use Microdata Sample (PUMS), from which we extract household-level housing, income, and demographic variables. The main outcome variable is the property value (VALP), which we log-transform for modeling. Key feature groups include:

- **Affordability and Income:** Household income (HINCP), family income (FINCP), income-to-value ratio, and income per person.
- **Structural Attributes:** Bedrooms (BDS), rooms (NR), household size (NP), and house age (derived from year built).
- **Location Context:** County and ZIP estimates via PUMA-to-ZIP crosswalks, used to link external neighborhood data.
- **Crime Context:** Aggregated violent and property crime counts from California county-level crime reports (2010–2022), merged by county and normalized to estimate relative safety.
- **Derived Features:** Rooms per person, income-to-value ratio, logarithmic transformed income, categorical safety levels and house age categories.

For a broader interpretability, we classify safety into ordinal risk tiers and encode housing maturity into categorical bins. Missing ZIPs and county codes limit spatial resolution in some records, especially for non-urban PUMAs.

To support exploratory modeling and visual diagnostics, we implement a custom Streamlit interface that enables users to:

- Visualize geographic coverage and model residuals using interactive maps.
- View feature importance (via SHAP), residual plots, and model performance comparisons.
- Inspect predictions for individual households and download filtered output.

This live dashboard integrates trained models, test data and the pre-processing pipeline as a reproducible front-end accessible at `portablepredictions.streamlit.app`. All feature engineering and model outputs are versioned and stored using `pkl` files. A sample visualization is shown in Figure 5.

3.3 Model Diagnostics and Residual Checks

For each model, we generate the following.

- Actual vs. predicted plots.
- Residual scatter and QQ-plots for normality and outliers.
- Residual histograms to spot systematic bias.
- Coefficient or feature importance plots.

These plots show consistent patterns: visible skewness, clustering, and banding, all pointing to missing local context in the data.

3.4 Model Interpretability with SHAP

To interpret the predictions of our best performing model (XGBoost), we use SHAP (SHapley Additive exPlanations), which decomposes the model output into additive contributions from each feature. For a given input $x_i \in \mathbb{R}^d$, the model prediction is expressed as:

$$f(x_i) = \phi_0 + \sum_{j=1}^d \phi_j^{(i)},$$

where ϕ_0 is the base value (mean prediction in the training set), and each $\phi_j^{(i)}$ represents the contribution of feature j to prediction $f(x_i)$.

We generate two main visualizations:

- **SHAP Summary Plot:** Shows the main features sorted by average impact, along with their direction and distribution of influence.
- **SHAP Dependence Plot:** Highlights interactions and nonlinear effects for key predictors (e.g., household income, crime rate).

These plots confirm that structural variables such as the number of bedrooms and household income play a significant role, but do not fully account for neighborhood-level price variation. Contextual features such as the safety score and crime rate appear prominently, underscoring the importance of geographic enrichment.

3.5 Known Limitations

Although our pipeline now runs end-to-end on California data, several limitations remain:

- The current implementation focuses on a single state (California); generalizing to other regions requires more robust geographic joins and consistent feature availability.
- PUMA-to-ZIP and ZIP-to-county crosswalks are still partially incomplete or mismatched, limiting spatial precision.
- Some key contextual drivers, including school quality, environmental risk, and parcel-level attributes such as lot size, are not yet integrated.

These constraints restrict the model’s ability to fully capture localized effects, leading to residual skewness and modest R^2 values across both linear and nonlinear models. Future phases will focus on expanding coverage to additional markets and enriching neighborhood-level features to support generalizable, interpretable prediction.

4 Preliminary Results

We trained four models, Linear Regression, Ridge Regression, Random Forest, and XGBoost, on ACS PUMS housing data enriched with structural and socioeconomic features and merged with California crime statistics. As shown in Figure 1, the ensemble models significantly outperform the linear baselines.

4.1 Prediction Accuracy

The XGBoost model achieved impressive $R^2 = 0.995$ and predicted most logarithmic transformation home values within tight bands of the true values, as shown in Figure 2.

4.2 Feature Importance

XGBoost’s built-in feature importance ranking (Figure 3) confirms that the income-to-value ratio dominates the prediction signal, followed by household income, family income, and crime-related variables such as violent crime rate and safety score.

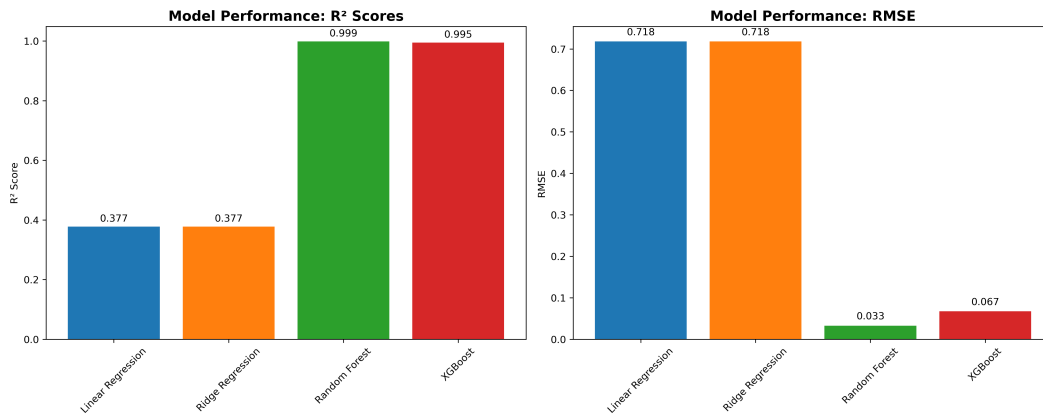


Figure 1: Model performance comparison: R^2 and RMSE across all four models. Random Forest and XGBoost outperform linear baselines.

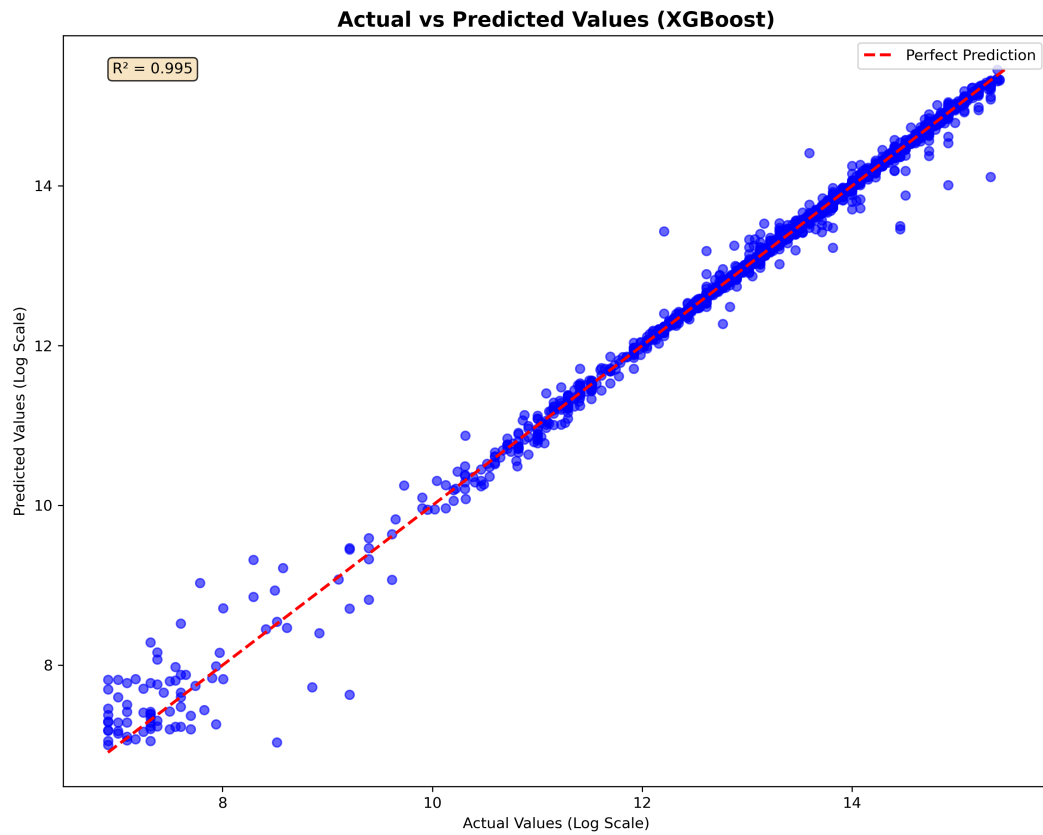


Figure 2: Actual vs. Predicted Values for XGBoost on log-transformed prices.

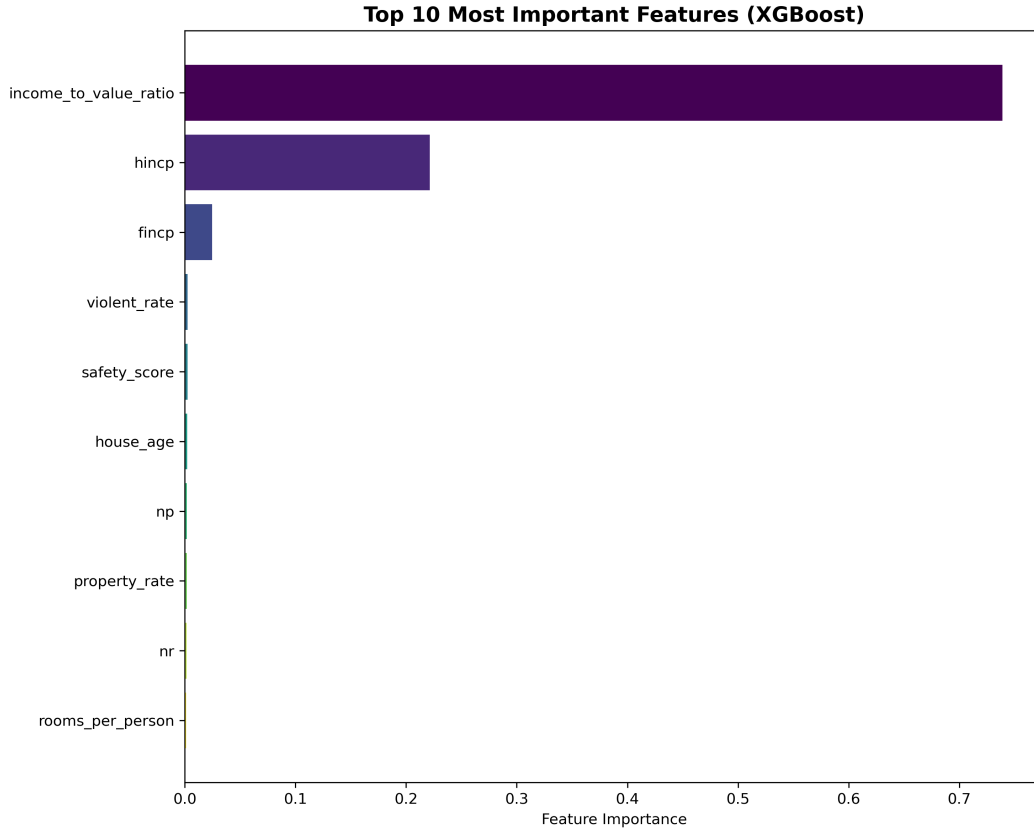


Figure 3: Top 10 most important features in the XGBoost model.

4.3 Residual Analysis

The residual plots in Figure 4 show minimal skewness and well-behaved variance in the ensemble models. Although minor outliers remain, residuals approximate a normal distribution centered around zero, suggesting a reasonable generalization.

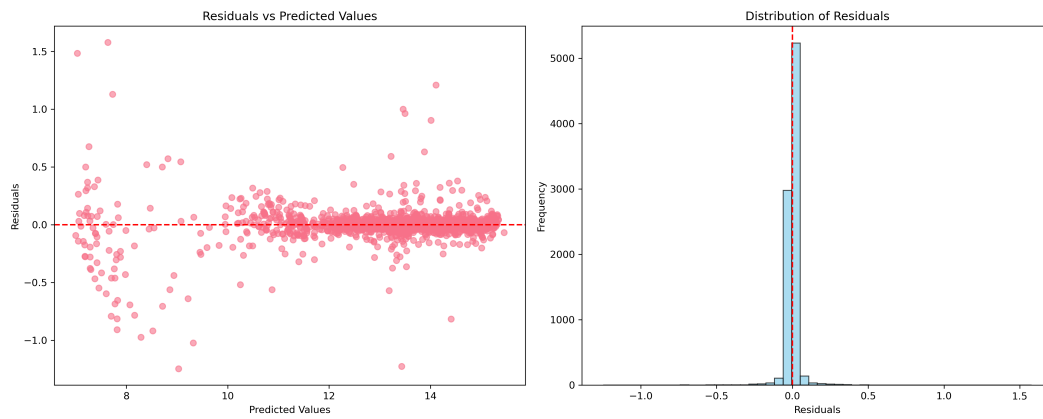


Figure 4: Left: Residuals vs. predicted values. Right: Histogram of residuals for XGBoost.

4.4 Geographic Visualization and Dashboard Insights

The Streamlit dashboard (Figure 5) provides interactive tools to explore prediction and model diagnostics geographically. The crime and value map helps surface ZIP-level inconsistencies and blind spots, guiding future data enrichment and model tuning.

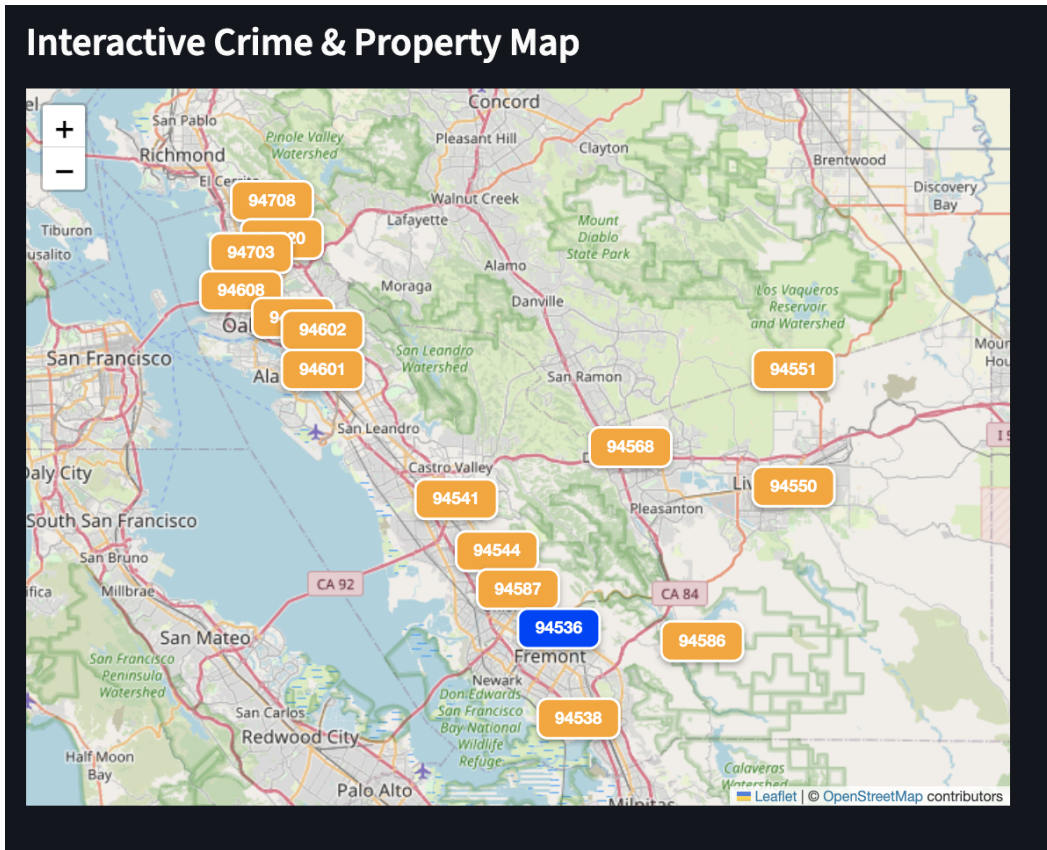


Figure 5: Interactive Crime & Property Map from the Streamlit dashboard.

4.5 Summary and Outlook

These early results demonstrate the effectiveness of ensemble methods in housing valuation, particularly when a socioeconomic and crime context is available. However, gaps in ZIP-to-PUMA crosswalks, neighborhood quality, and real estate characteristics such as lot size still limit broader generalization. Future work will focus on completing spatial joins, enriching contextual features, and expanding coverage to test geographic portability.

4.6 Next Steps

Our new pipeline has progressed considerably from previous baselines, adding California crime data and enhancing model diagnostics using a scalable Streamlit interface. However, there are a number of areas that need to be improved in the future.

First, although XGBoost and Random Forest yielded high R^2 values, the classification of feature importance and residual diagnostics suggest that the models are still very sensitive to missing or out-of-place geographic context. The income-to-value ratio dominates the prediction, but in the absence of decent ZIP-level attributes, local variation remains unexplained.

Second, geographic generalization to states outside California has not been tested. To assess model portability, we plan to generalize the pipeline to a number of states through additional ACS PUMS extracts. This will involve completing and confirming the PUMA $\rightarrow$ ZIP $\rightarrow$ County mapping layer and increasing how the crime and neighborhood context are spatially joined.

Third, feature augmentation remains relevant. We intend to add school quality measures, zoning classifications, mortgage status, and environmental attributes (flood zones, say) to better place housing prices in context. These additions will also allow us to better understand and discern spatial heterogeneity in SHAP output.

Finally, systematic hyperparameter tuning of XGBoost and Random Forest - the algorithms we are focusing on - will be done through cross-validation and grid/randomized search. This ensures our ensemble models are not under-optimized and stabilizes performance across regions.

Ultimately, we want to have an open, reproducible, and regionally adaptive AVM pipeline that can serve decision-making in housing, investment, and policy.

5 Future Plans

Our early results show that improving housing price prediction is not just a matter of model selection — it is fundamentally about enhancing the spatial and contextual quality of the data. The next phase will focus on strengthening geographic joins, enriching neighborhood-level features, and optimizing the modeling pipeline for generalization and interpretability. Specifically, we plan to:

- **Finalize ZIP Crosswalks and Spatial Joins.** Our current geographic joins rely on partial PUMA-to-ZIP mappings. These will be cleaned up and finalized to support the integration of external datasets such as school quality, zoning, and transit proximity. Reliable spatial linkage is essential to incorporate location-based context and improve prediction accuracy.
- **Expand Contextual Features.** To address persistent residual bias and spatial clustering, we will integrate richer neighborhood attributes: crime rates at ZIP or census tract level, school district rankings, zoning types, proximity to green spaces and accessibility to transit. This move beyond structural variables aims to improve both predictive performance and interpretability.
- **Refine Data Quality.** We observed banding and skew that point to inconsistencies in the underlying data. We will conduct a detailed data audit, checking for missing or mismatched geographic codes, anomalous income or value entries, and inconsistent joins, to ensure clean, reliable inputs before retraining.
- **Tune and Validate Models.** Hyperparameter tuning for Random Forest and XGBoost will be done using cross-validation, with performance evaluated across multiple regions. This will allow us to test the generalizability and stability of the results beyond California. Diagnostics such as residual plots, SHAP distributions, and accuracy heatmaps will be re-run post-tuning.
- **Enhance Crime Data Resolution.** Current crime estimates are based on county-level aggregates. We aim to improve these with incident-level data from public sources (e.g. <https://www.crimemapping.com>) and local law enforcement agencies, geocoding reports to ZIP or tract level where available.
- **Add Proximity Features.** We will derive distance-based variables measuring proximity to schools, transit stations, parks, and commercial hubs using GIS techniques. These features are key to capturing neighborhood desirability and should help reduce unexplained variance in pricing models.

Together, these efforts aim to transform the project from a strong technical prototype to a regionally adaptable, interpretable, and trustworthy tool for housing analysis and decision-making.

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